WASP-2b

R-0578

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_230831` · created 2026-07-24T17:29:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460187.7693 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.095 +/- 0.046
Transit depth (Rp/Rs)^2	0.91 +/- 0.87 [%]
Orbital Inclination (inc)	80.07 +/- 2.86
Airmass coefficient 1 (a1)	0.046 +/- 0.01
Airmass coefficient 2 (a2)	2.67 +/- 0.98
Scatter in the residuals of the lightcurve fit is	19.43 %
Best Comparison Star	#2 - [326, 323]
Optimal Aperture	1.42
Optimal Annulus	10.38
Transit Duration (day)	0.011 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.095 ± 0.046 vs 0.1326 (deviation 0.82σ)
Duration fit vs published	0.011 d vs 0.0737 d (deviation 2.61σ)
Mid-time O–C	9.9 min
Depth SNR	2.1
Residual scatter	19.43 %

Light curve

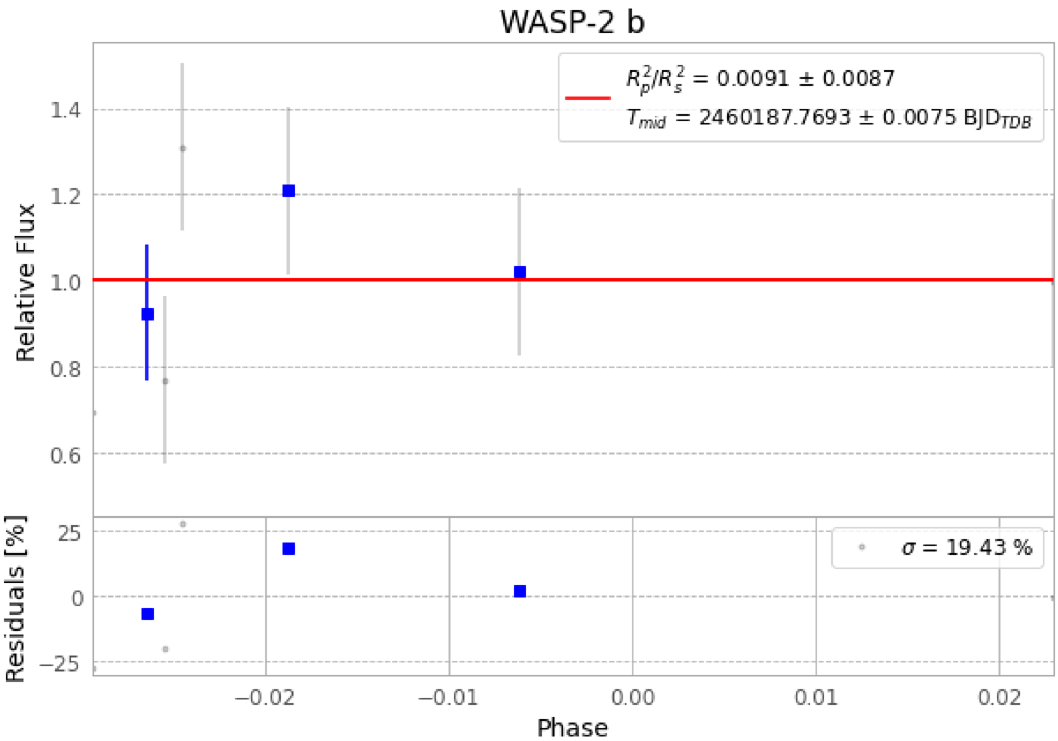

Figure 1 — FinalLightCurve_WASP-2 b_2023-08-30.png (SHA-256 6cae8cebbbe876b1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 119], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6dd64bbf5e7bbecf…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

72 FITS frames (47 MB), `WASP-2230831043315.FITS` → `WASP-2230831080615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-230831.log (a002fc27c55c…), FinalLightCurve_WASP-2 b_2023-08-30.png (6cae8cebbbe8…), FinalLightCurve_WASP-2 b_2023-08-30.csv (9a0aa7f6da0c…)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)